Dream Paris Wedding

AI Communication Assistant

Technical Brief — Developer Team

1. Project Overview

Dream Paris Wedding is embarking on a long-term initiative to build a custom AI-powered assistant designed to handle communication management, message drafting, and workflow automation across all major channels. The goal is to preserve the brand's high-touch, personalised communication style while freeing the team to focus on high-value planning and client relationships.

1.1 Core Objectives

- Automate drafting of routine and repetitive messages across Gmail, WhatsApp Business, and Instagram
- Classify all incoming messages by category for priority-based inbox management
- Maintain Dream Paris Wedding's exact brand voice, tone, and communication standards
- Implement a human-approval workflow for sensitive or high-value client interactions
- Provide a clean, easy-to-use dashboard for reviewing and approving AI-drafted replies
- Build a scalable, long-term infrastructure — not a temporary automation patch

1.2 What the System Will NOT Do

- Replace human judgment for complex or emotionally sensitive client conversations
- Send responses autonomously for high-value inquiries without team review
- Operate as a public-facing chatbot or self-service widget

2. Description

The following is a plain-language breakdown of the five core components that make up the AI assistant system:

A. Smart Inbox Reader

The system connects to Gmail, WhatsApp Business, and Instagram DMs automatically. It reads incoming messages, identifies the sender, and understands the context — whether this is a new inquiry, existing client, or vendor — without any manual sorting.

B. Context-Aware AI Drafter

Using OpenAI's GPT-4 API, the assistant analyses each message and generates a reply written in Dream Paris Wedding's own voice. Drafts are produced from a custom knowledge base containing past email examples, standard responses, and brand communication guidelines.

C. Approval-Based Sending Logic

Not every message gets auto-sent. The system applies decision rules to determine whether a drafted reply is sent automatically or placed in a pending approval queue for a team member to review. High-value inquiries and sensitive discussions always require human sign-off.

D. Message Classification Engine

Every incoming message is automatically tagged into one of five categories: New Inquiry, Existing Client, Vendor Communication, Collaboration Request, or General Question. This drives inbox organisation and response prioritisation.

E. Custom Knowledge Base

The AI's tone and accuracy depend entirely on this knowledge base, which is populated with real email examples, brand guidelines, lead qualification scripts, vendor templates, and planning process documentation.

3. Technology Stack — Why n8n

After careful evaluation of all available development approaches, the system will be built using n8n — a powerful open-source workflow automation platform — as the core engine, combined with a custom React dashboard for the approval interface.

3.1 Technology Components

Layer	Technology	Purpose
Workflow Automation	n8n	Core engine — reads messages, calls OpenAI, routes approvals, sends replies
AI Intelligence	OpenAI GPT-4o API	Message classification, draft generation, tone matching
Email	Gmail API via n8n	Read inbox, create drafts, send approved replies
WhatsApp	Meta Cloud API via n8n	Receive and send WhatsApp Business messages
Instagram	Meta Graph API via n8n	Read and reply to Instagram DMs
Approval Dashboard	ReactJS + Node.js	Web interface for reviewing and approving AI drafts
Database	MongoDB	Store messages, drafts, approval history, and analytics
Hosting	VPS Server (self-hosted)	Single server hosts n8n + all client workflows
Knowledge Base	JSON / Vector store in n8n	Brand guidelines, past emails, response templates

4. How the System Works — Message Flow

Step	What Happens	Technology
1. Receive	System polls Gmail, WhatsApp & Instagram every few minutes for new messages	n8n trigger nodes
2. Classify	GPT-4 reads the message and assigns a category and priority level	OpenAI node in n8n
3. Retrieve	Relevant templates and past emails are pulled from the knowledge base	n8n + knowledge base
4. Draft	GPT-4 generates a reply using brand tone and retrieved context	OpenAI node in n8n
5. Route	Decision rules check — auto-send or send to approval queue?	n8n conditional logic
6. Review	Team sees original message + AI draft side by side in the dashboard	React dashboard
7. Approve	One click approves — system sends reply via the correct channel API	n8n + Gmail/Meta APIs
8. Log	All messages, drafts, and replies are logged for review and improvement	MongoDB

5. The Approval Dashboard

A custom web dashboard is included as part of the system. This is the primary interface your team uses daily to review AI-drafted replies, manage the approval queue, and monitor performance across all three communication channels.

5.1 Dashboard Features

- Side-by-side view of original message and AI-generated draft reply
- One-click approve, edit, or discard for each draft
- Message category tags — New Inquiry, Existing Client, Vendor, Collaboration, General
- Sending mode selector per message — Auto, Approve First, or Draft Only
- Real-time stats — messages pending, auto-sent, approved, average response time
- Live channel activity bars for Gmail, WhatsApp, and Instagram
- Full message history and audit log

5.2 Dashboard Preview

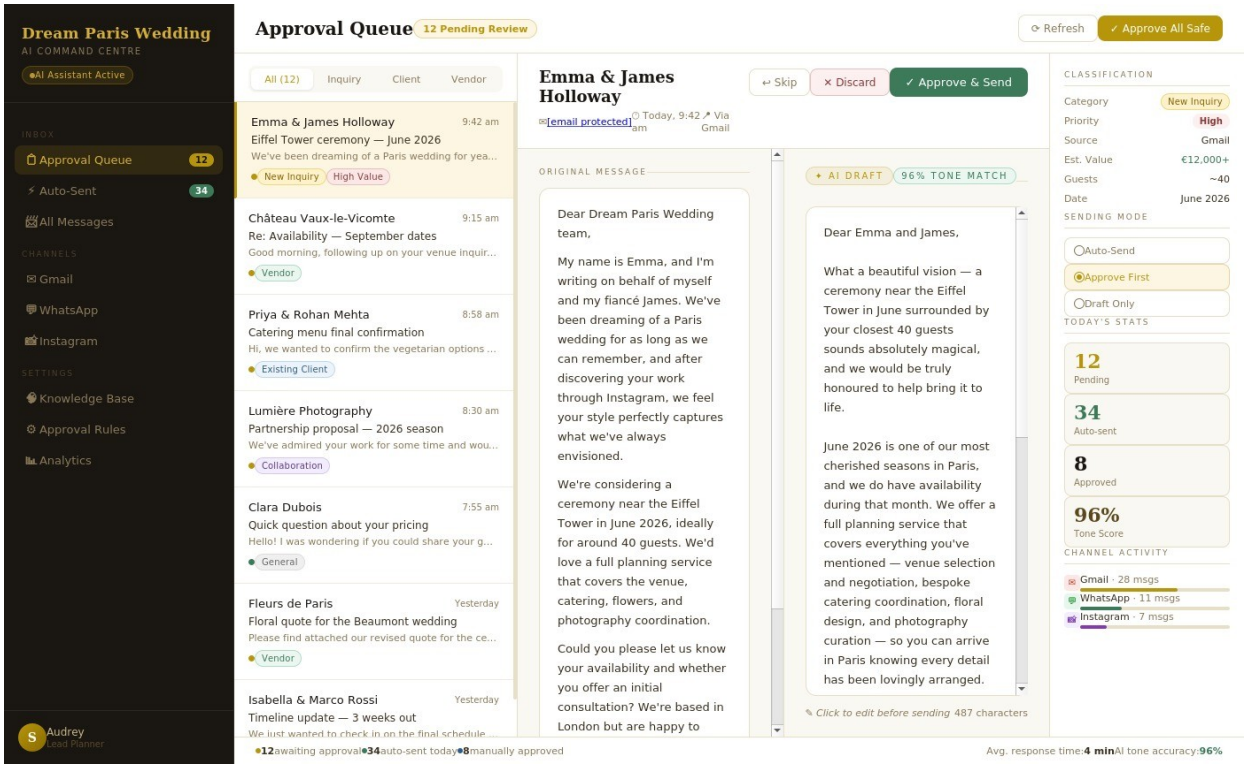

Figure 1 — Dream Paris Wedding AI Assistant Dashboard

6. Message Classification & Handling Rules

Category	Examples	Suggested Handling
New High-Value Inquiry	Wedding date, venue questions, full planning requests	ALWAYS human approval before reply
Existing Client	Timeline updates, vendor confirmations, changes	Draft shown to team; auto for simple follow-ups
Vendor Communication	Caterer intros, florist quotes, logistics	Semi-automated with standard templates
Collaboration Request	Photographer partnerships, venue referrals	Draft for human review
General Question	Pricing, availability, location queries	Can be auto-replied with FAQ responses

7. Building the Knowledge Base

The knowledge base is the most important element of the entire project — it is what gives the AI its voice. Without high-quality, structured content, the AI will produce generic, off-brand replies. The following content must be compiled before development begins:

- **Real Examples:** Past sent emails — minimum 50 to 100 examples across all categories
- **Templates:** Standard reply templates for the most common 10 to 15 inquiry types
- **Tone Guide:** Brand voice guidelines — phrases you use, phrases you avoid, emotional tone
- **Qualification:** Lead qualification questions — what you always ask new clients
- **Process Docs:** Planning process documentation — onboarding steps, timeline, milestones
- **Vendor Tone:** Vendor communication style — formality level, standard greetings
- **FAQs:** Pricing and availability FAQs — answers to the most repeated questions

8. Phased Implementation Roadmap

The system is built in structured phases to ensure Dream Paris Wedding has a working assistant as quickly as possible, with each phase adding capability on top of a proven foundation.

Phase	Timeline	What Gets Built	Milestone
Phase 0 Foundation	Week 1–2	Knowledge base compilation, brand voice guide, email library, Meta API application submitted	Content ready for AI training
Phase 1 Gmail MVP	Week 2–4	n8n installed on server, Gmail connected, OpenAI drafting live, basic classification, approval dashboard v1	Gmail assistant live and in use
Phase 2 Refinement	Week 5–6	Knowledge base expanded from real usage, tone fine-tuned, auto-send rules activated for low-risk categories, dashboard v2	System running with confidence
Phase 3 WhatsApp	Week 7–8	WhatsApp Business API integrated into n8n, message drafting extended to WhatsApp, unified queue	Two-channel system live
Phase 4 Instagram	Week 9–10	Instagram DM integration, all three channels unified in one dashboard, full channel activity view	Full three-channel system live
Phase 5 Optimisation	Ongoing	Monthly knowledge base updates, tone reviews, performance analytics, new template additions	Continuous improvement

9. API Setup & Credentials Required

Before development begins, the following API credentials and accounts must be created and configured. Each developer should have access to the shared credentials vault.

API / Service	Account Required	Where to Register	Notes
OpenAI API	OpenAI Platform account	platform.openai.com	Generate API key — use GPT-4o model
Gmail API	Google Cloud project + OAuth 2.0	console.cloud.google.com	Enable Gmail API, configure OAuth consent screen
WhatsApp Business API	Meta Business Manager account	business.facebook.com	Requires phone number + business verification — apply immediately
Instagram Messaging API	Meta Developer App + Instagram Professional Account	developers.facebook.com	Requires App Review — allow 1–4 weeks
n8n	Self-hosted on VPS	n8n.io/cloud or own VPS	Install via Docker on server
MongoDB	MongoDB Atlas (cloud) or self-hosted	mongodb.com/atlas	Free tier sufficient for Phase 1
Server / VPS	Any VPS provider	DigitalOcean / Hetzner / AWS	Minimum 2GB RAM, Ubuntu 22.04

10. Developer Responsibilities

Developer	Primary Responsibilities	Tools
Abdur Rehman	n8n workflow design and all API integrations (Gmail, WhatsApp, Instagram, OpenAI), MongoDB schema, Node.js backend APIs	n8n, Node.js, MongoDB
Usama	React approval dashboard frontend, dashboard-to-backend integration, UI components, testing interface	ReactJS, n8n (support)
Both	Knowledge base configuration, end-to-end testing, deployment, client training session	n8n, Cursor / Claude AI

10.1 Development Guidelines

- Use Cursor or Claude to accelerate code generation — all API integration modules should be AI-assisted
- Build n8n workflows as reusable templates from Day 1 — each workflow must be easily duplicated for new clients
- All API credentials must be stored as environment variables — never hardcoded in workflow or code
- Each phase must be tested end-to-end before moving to the next phase
- Maintain a shared changelog document — log every workflow change, update, and fix
- Dashboard UI must work on both desktop and mobile browsers
- All n8n workflows must be exported as JSON and version-controlled after each major update

11. Technical Risks & Mitigations

Risk	Likely Cause	Mitigation
Meta API approval delays	WhatsApp/Instagram App Review takes 1–4 weeks	Submit Meta application on Day 1 before writing any code
Gmail OAuth token expiry	Google OAuth tokens expire after inactivity	Implement token refresh logic in n8n — test regularly
OpenAI API rate limits	High message volume hitting API limits	Implement retry logic and queue system in n8n workflow
n8n workflow breaking on update	n8n version update changes node behaviour	Do not auto-update n8n — test updates on staging before applying to production
Knowledge base producing off-tone replies	Insufficient or poor quality training examples	Minimum 50 real email examples required before Go Live — do not skip
Server downtime affecting all clients	Single point of failure on shared VPS	Set up uptime monitoring (UptimeRobot) and auto-restart scripts
Dashboard authentication security	Unsecured admin access	Implement JWT auth on dashboard — never deploy without login protection

12. Developer Action Plan

The following tasks should be completed in order before writing any workflow or application code:

1. **Server setup** Provision a VPS (minimum 2GB RAM, Ubuntu 22.04). Install Docker, n8n, and MongoDB.
2. **API credentials** Create OpenAI account and generate API key. Set up Google Cloud project and enable Gmail API with OAuth 2.0.
3. **Meta application** Submit WhatsApp Business API and Instagram Messaging API application immediately — this is the longest lead time item.
4. **n8n master workflow** Build the master workflow template in n8n — Gmail trigger → OpenAI classify → OpenAI draft → route to approval queue. This is the core engine everything else builds on.
5. **Dashboard scaffold** Usama to set up React project with routing, auth, and basic layout. Connect to Node.js backend API.
6. **End-to-end Phase 1 test** Run a complete test: send an email to the Gmail inbox → watch it appear in the approval dashboard → approve it → confirm it sends from Gmail. Only move to Phase 2 once this works reliably.
7. **Knowledge base load** Once client provides email examples, load them into the knowledge base and test tone accuracy before any live use.

Build each phase to production quality before moving to the next. A reliable Gmail assistant delivered in 3 weeks is worth more than a broken three-channel system delivered in 10.